

PicoCTF 2026 – Binary Digits

Binary Digits

Forensics

Easy

100 pts

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This file doesn't look like much... just a bunch of 1s and 0s. But maybe it's not just random noise. Can you recover anything meaningful from this?

Download the file [here](#) ↗.

In this challenge, they want us to recover a file that is all ones and zeros

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Binary to ASCII Conversion

```
—$ cat digits.bin  
11111111110110001111111111110000  
000000000000000000000100000000000000  
0110000001110000011000000101000
```

When we look at the contents of the file, it is indeed only ones and zeros

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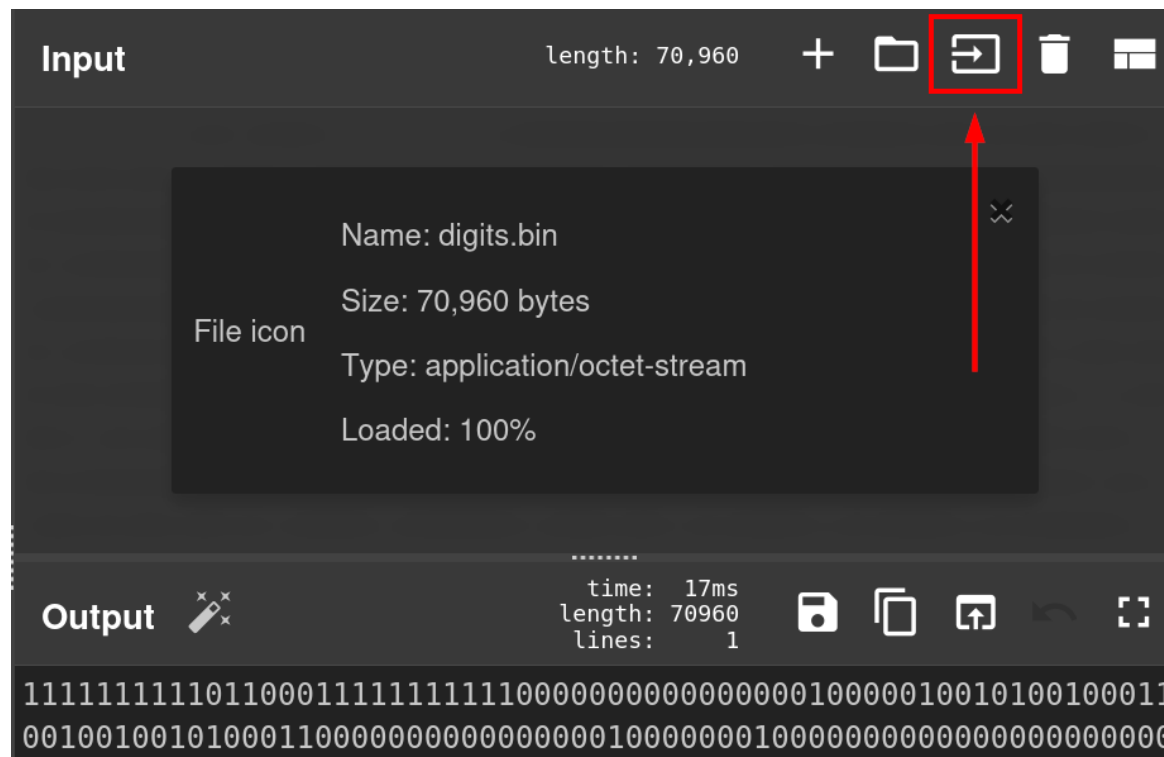
Binary to ASCII Conversion

ASCII Conversion	
Binary	11111111
Hexadecimal	FF
ASCII	Unprintable

If we treat each 8 binary digits as an ASCII character, we can convert the binary digits into a file

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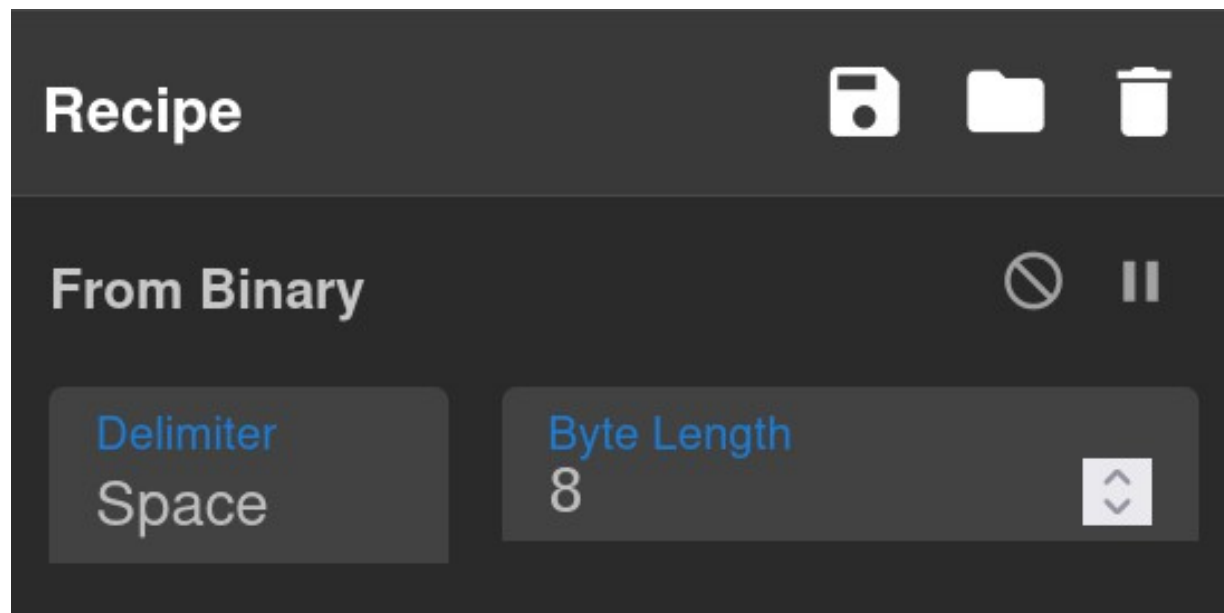
CyberChef Solve



We can solve this in CyberChef by importing the file

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CyberChef Solve



Then apply the “From Binary” operation, which converts binary numbers to ASCII bytes

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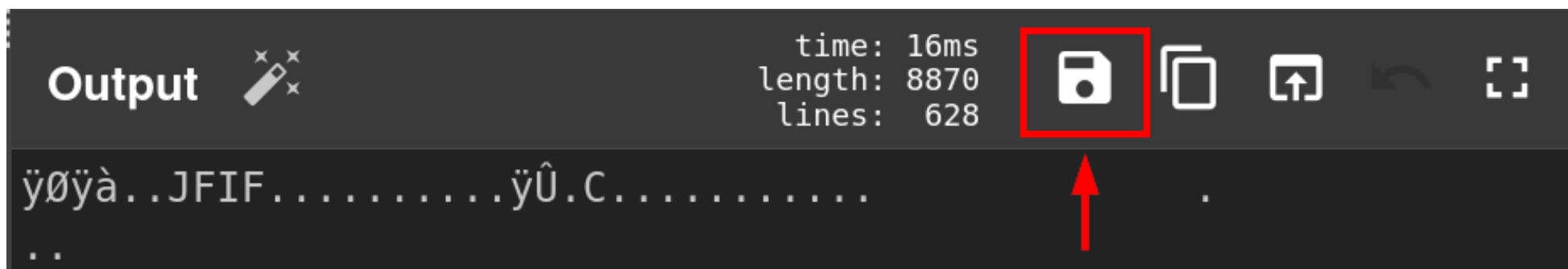
CyberChef Solve

```
Output ✎
ÿøÿà...JFIF.....ÿÛ.C...
..
.....$. ' ",#...
(7),01444.'9=82<.342ÿÛ.C.
```

The file contents reveal that this is a picture file
(JPG)

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CyberChef Solve



We can save this output to a file, and give it a .jpg file extension

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CyberChef Solve

A screenshot of a CyberChef interface. The top bar shows a file named 'picoCTF[Binary_Digits_20260315]' in red text. The main area is empty, suggesting the file has been loaded but no operations have been applied yet.

picoCTF[Binary_Digits_20260315]

We can then open the file and retrieve the flag

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Python Solve

```
data = "".join(data.split())  
output = bytes(int(data[i:i+8], 2) for i in range(0, len(data), 8))
```

To solve this challenge with Python scripting, we need to convert the data into a bytes from each 8-digit chunk